AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM291

Management Information Systems

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Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 20/4/2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 1.5 Hours

Instructions:

- Total Marks: **30**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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Answer All the following questions:

Question 1: Select the correct answer (10 Marks)

1. Businesses use information systems primarily to achieve which business objective?
   a) Survival
   b) Operational excellence
   c) new products and services
   d) All the above
2. What is the correct definition of Data?
   a) Data shaped into a meaningful and useful form
   b) Summarized reports for middle management
   c) Streams of raw facts
   d) Information that has been processed and analyzed
3. Google Drive is:
   a) Virtual meeting systems
   b) Enterprise social networking tools
   c) Cloud collaboration services
   d) Email and instant messaging (IM)
4. Which group is primarily served by Transaction Processing Systems (TPS)?
   a) Senior Management
   b) Middle Management
   c) Operational Management
   d) Scientists and knowledge workers
5. A manager asking "What-if" questions about production impacts is likely using
   a) Transaction Processing System (TPS)
   b) Decision Support System (DSS)
   c) Enterprise Content Management (ECM)
   d) Knowledge Management System (KMS)
6. Enterprise Resource Planning (ERP) systems are designed to:
   a) Manage relationships specifically with external suppliers
   b) Integrate data from key business processes into a single system
   c) Coordinate social networking platforms for employees
   d) Focus exclusively on the sales and marketing functional area
7. Which activity is found in Problem Identification?
   a) Testing a chosen solution
   b) Agreeing that a problem exists and defining its causes
   c) Measuring the outcomes of a new system
   d) Evaluating the cost and feasibility of alternatives
8. What is used to clarify table relationships in a relational database?
   a) Many-to-many relationship
   b) Collection of related files
   c) Entity-relationship diagram
   d) Record
9. What is the purpose of a Primary Key?
   a) To link two tables together in a join
   b) To store the specific characteristics of an attribute
   c) To uniquely identify each record in a table
   d) To minimize the need for normalization
10. Within ESG, the "Governance" pillar primarily relates to:
    a) Carbon footprint and energy use
    b) Executive pay, internal controls, and shareholder rights
    c) Relationship with the community and recycling practices
    d) Diversity and inclusion in the supply chain only

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Question 2: Business Drivers and Systems Foundations (6 Marks)

A. Identify and briefly explain three of the six core business objectives that drive firms to invest in information systems. (3 Marks)

B. Differentiate between Data and Information. Provide one specific example of how a business might transform raw data into meaningful information. (2 Marks)

C. Define Feedback within the context of an information system and explain its role in helping an organization. (1 Mark)

Question 3: Critical Thinking and Problem-Solving (8 Marks)

Scenario: A regional retail chain is experiencing a significant drop in customer retention. Management suspects that the current manual system for tracking customer complaints and loyalty points is too slow and prone to errors.

A. Using the Four-Step Problem-Solving Process, describe the actions the retail chain should take in the "Problem Identification" and "Solution Design" stages. (3 Marks)

B. Analyze this scenario through the Three Dimensions of Information Systems. Identify one potential Organizational cause, one Technology cause, and one People cause for the retention problem. (3 Marks)

C. Explain the role of Critical Thinking in this process. Why is "suspending judgment" important when evaluating multiple technology alternatives? (2 Marks)

Question 4: Enterprise Applications and Business Processes (8 Marks)

A. Describe the difference between a Transaction Processing System (TPS) and a Decision Support System (DSS) in terms of the management level they serve and the types of decisions they support. (2 Marks)

B. What is a Cross-Functional Business Process? Provide an example of how an Enterprise Resource Planning (ERP) system facilitates such a process across functional areas like Sales and Finance. (3 Marks)

C. Companies are increasingly striving for ESG Leadership. Explain how an information system can help a company achieve its "Environmental" and "Governance" goals. (3 Marks)

Question 5: Database Design and Information Management (8 Marks)

A. Illustrate the Data Hierarchy by arranging the following from the smallest to the largest level: *Database, Bit, Record, Byte, File, Field.* (2 Marks)

B. You are tasked with designing a simplified relational database for a library.

1) Identify two Entities relevant to a library and list two Attributes for each. (2 Marks)
2) Define a Primary Key and a Foreign Key and explain their role in maintaining Referential Integrity between your library tables. (2 Marks)

C. Explain the concept of Normalization. Why is it necessary to "streamline" complex groups of data into two-dimensional tables? (2 Marks)

End of questions

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